

MAINTENANCE PROCEDURES - 11 kV SWITCHGEAR

Scope: routine and corrective maintenance for the Ring Main Unit (RMU) and the Circuit Breaker (CB).

ROUTINE INSPECTION OF THE RING MAIN UNIT

WARNING: Never open the RMU cover while the busbar is live. Confirm isolation with the substation in charge.

1. Isolate the Ring Main Unit (RMU) and apply the lockout tag.
2. Verify absence of voltage with the approved voltage detector.
3. Clean the bushings with a lint free cloth.
4. Check the SF6 gas pressure gauge reads between 1.2 bar and 1.5 bar.
5. Tighten the earth connection bolt to 40 Nm.
6. Record the operation counter reading in the log book.

REPLACING THE MAIN CIRCUIT BREAKER (CB) CONTACTS

CAUTION: Contacts may retain heat. Allow 30 minutes cooling before handling.

1. De-energise the panel and earth both sides.
2. Remove the arc chute assembly, part number P/N 88123-A.
3. Replace the fixed and moving contacts as a matched set.
4. Apply contact grease sparingly to the silver plated surfaces.
5. Torque the contact stem nut to 25 Nm.
6. Measure contact resistance; it must be below 60 micro-ohm.

SIX MONTHLY MAINTENANCE OF THE AUXILIARY TRANSFORMER

WARNING: Insulation testing injects high voltage. Clear all personnel before starting.

1. Switch off the Miniature Circuit Breaker (MCB) feeding the auxiliary transformer.
2. Measure insulation resistance at 1 kV; the value must exceed 100 megaohm.
3. Check the cooling fan current draw, rated 1.8 A.
4. Verify the oil level sight glass is between the minimum and maximum marks.
5. Torque the HV terminal bolts to 55 Nm.